The Drug Main Effect in Aggregated N-of-1 Designs under the Refined Process: ADEMP Working Protocol for the Rebuild of Paper 04

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Manuscript. `analysis/report/04-treatment-main-effect/report.Rmd` (to be rebuilt).

Protocol history. This file replaces the working protocol of 2026-05-07, which is retained unchanged as `analysis/scripts/treatment-main-effect/00-ademp-protocol-v1.md` (byte-identical copy, checked with `cmp`). Version 1 describes the study behind the current manuscript and its driver `production-run.R`; it is kept as the record of that study.

Purpose. This working protocol sets out a new simulation study of the treatment (drug) main effect β_D in aggregated N-of-1 designs, to replace the study of the current paper 04. It is a planning document: no code has been written for it and nothing has been simulated. Section 8 lists the decisions that must be settled before code is written. The study is exploratory. Its criteria may be revised as results come in, every revision will be recorded as an amendment with its reason, and its findings should be confirmed in an independent study. The structure follows ADEMP: aims, data-generating mechanisms, estimands, methods and performance measures (Morris, White and Crowther [1]).

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here and are not part of the canonical set.

Table 1: Notation index for symbols used in this protocol

Symbol	Meaning	Status
i, t	participant and visit indices; $t = 0$ is baseline	canonical
w_t	calendar week of visit t	local (`docs/40`)
N	total participants across all paths or arms	canonical
P	number of randomization paths (4 Hybrid, 2 OL+BDC, 2 CO)	canonical
Y_{it}	symptom score (CAPS points; code column `Sx`)	canonical
BL_i	baseline level	canonical
BR_{it}, PB_{it}, TV_{it}	drug, placebo-belief and natural-course responses	canonical
η_t	design expectancy: 1 open label, 0.5 blinded, 0 at baseline (code `De`)	canonical
D_{it}, $D_{bc,it}$	binary drug state; exposure-decayed indicator, 0 before any exposure (code `Dbc`)	canonical
t_{od}, t_{sd}, t_{pb}	time on drug, time since discontinuation, time under expectancy	canonical
$t_{1/2}$, $t^a_{1/2}$	true carryover half-life; half-life assumed by an analysis (weeks)	canonical; local (`docs/45`)
$G(t; m, d, r)$	modified Gompertz curve	local (`docs/40`)
m_{BR}, m_{TV}, m_{PB}	Gompertz ceilings of the three components (published 10.986, 6.506, 6.506)	local
f_m	drug-effect multiplier, $m_{BR}/10.986$	local
q_{it}	unit drug-response profile: the BR mean of participant i's path divided by m_{BR}, anchored carryover included	local
v_i	participant-specific deviation of the drug-response ceiling	local
τ^2	variance of v_i, the participant-by-treatment variance (σ^2_v of `docs/51`)	local
β_D	drug main effect, the primary estimand	canonical
ρ, c_1	AR(1) serial correlation per week (0.7); cross-component correlation (0.2)	canonical; local (`docs/40`)

Symbol	Meaning	Status
σ_{BR}	SD of BR (8)	canonical
$\sigma_{\varepsilon,0}$, r_{BL}	SD of measurement error added to the observed baseline; its reliability $\sigma^2_{BL}/(\sigma^2_{BL}+\sigma^2_{\varepsilon,0})$	local
$\sigma_{\varepsilon,\mathrm{OL}}$, $\sigma_{\varepsilon,\mathrm{bl}}$	SD of measurement error at open-label and blinded visits	local (`docs/46`)
κ	mean model SE over empirical SD	canonical
α, π	nominal level (0.05); power	canonical
n_{reps}	Monte Carlo replicates per cell	canonical

Two collisions are avoided deliberately. `docs/37` uses τ^2 for the residual variance of the paired contrast given the biomarker; that quantity is written $\sigma^2_{\Delta|B}$ here, and τ^2 is reserved for the participant-by-treatment variance, as in the literature on series of N-of-1 trials [2, 3]. The canonical δ (standardized bias) and θ (generic interaction coefficient) are not used for the drug effect.

Sign convention (NOTATION.md, rule 5). The components are non-negative reductions in symptom severity, so an increase in any of them lowers Y. The drug main effect β_D is therefore negative when the drug helps; m_{BR}, f_m and τ are non-negative. Bias is reported on the β_D scale, so a positive bias is a bias toward no effect.

Units. Carryover half-lives are in **weeks** throughout, matching the refined process and the interaction manuscripts. This departs from `NOTATION.md`, Part 1, "Units", which assigns days to the main-effect manuscript; that paragraph must be revised when the rebuilt paper is written (1 week = 7 days). Effect sizes are given in CAPS points and in SD units (Section 3.5).

Evidence labels. As in `docs/53`: *quoted*, *inspected* (read from source), *derived* (follows algebraically), *verified* (computed by running code; here only arithmetic checks in the session scratchpad, `estimand-check.R`, not a project script), *inferred* and *assumed*.

Glossary.

- **The refined process.** The recommended data-generating process (DGP) of paper 01, Table `tab:spec`; "Refined" in tables. **The published process**: Hendrickson's code at commit 58b32a9 (`NOTATION.md`, "Named data-generating processes").
- **Hybrid.** The Hendrickson design: open label to week 8, blinded discontinuation in weeks 8 to 12, a blinded two-period crossover in weeks 12 to 20; visits at weeks 4, 8, 9, 10, 11, 12, 16, 20 [4].
- **Multi-crossover.** The design of the current paper 04: eight one-week periods, four on drug and four on placebo in a random order per participant. It is not the Hybrid design.

- **Drug-expectancy confounding.** In open-label phases every visit is on drug at full expectancy, so the drug state and η_t are correlated (0.588 over the nine Hybrid rows; `compendium-mathematics.tex`, Section "The published Hybrid construct", verified there). An analysis that does not model η_t credits part of PB to the drug.
- **Heterogeneous null.** A cell with $f_m = 0$ and $\tau > 0$: the mean drug effect is zero but individual effects are not.
- **Size-adjusted power, shortfall, minimax regret.** As in `docs/46` and `docs/47`: power at the 5% quantile of the analysis's own null p-values; an analysis's shortfall in a cell is its distance from the best surviving analysis there; the selected analysis minimizes its largest shortfall.
- **Oracle analysis.** An analysis that uses information the trial would not have (a true half-life, the true Gompertz shape, an error-free baseline). Reported for reference; never selectable.

1. Why the rebuild

The current paper's simulation does not study the design, the process or the inference that the paper describes. Seven problems were found (inspected, 2026-10-08):

1. **The design is not the Hybrid design.** The driver generates a randomized 4 + 4 multi-crossover of one-week periods (`production-run.R`, lines 80 to 85: `seq_base` of four 1s and four 0s, permuted by `sample()`), while the abstract and Results call the N-of-1 arm "hybrid" (`report.Rmd`, line 8; lines 1216 to 1221). Methods describes the multi-crossover correctly (lines 515 to 517).
2. **The outcome has no placebo response, natural course, expectancy or onset.** The outcome is `baseline + Dbc * true_effect + carryover_adj + eps` (`production-run.R`, line 103). There is no PB, no TV, no η_t and no Gompertz onset, so the drug-expectancy confounding of the open-label phase never arises, and the analysis's linear time term is fitted to a mean with no trend.
3. **No treatment-effect heterogeneity.** Every participant has the same effect (`true_effect`, line 103); the paper says that between-patient heterogeneity lies "purely" in baseline frequency (`report.Rmd`, lines 1362 to 1374), although its own literature review rests on τ^2 (lines 153 to 252).
4. **Carryover exclusion uses oracle information.** The analysis drops the rows flagged `is_carryover`, a column the DGP writes (`production-run.R`, lines 110 and 120 to 127). A trial does not know which visits carry residual drug effect.
5. **Carryover is truncated and on a day scale.** Carryover reaches only the period immediately after a drug period (lines 94 to 101), and half-lives are in days (lines 53 and 95), off the scale of every other paper in the compendium.

6. **The inference labeled Kenward-Roger is not.** The test uses the `nlme` table's degrees of freedom (lines 172 to 178), which are the containment (between-within) degrees of freedom, not Kenward-Roger and not Satterthwaite, although the header (line 13) and Methods (`report.Rmd`, lines 564 to 568) say otherwise; the reduced model fitted for the correction (lines 159 to 170) is never used.
7. **The randomized-trial covariate is measured without error.** The RCT endpoint is generated from, and adjusted for, the participant's true intercept (`production-run.R`, lines 197 to 204 and 235). An ANCOVA on an error-free covariate removes all between-participant baseline variance, which favors the RCT in a way no trial can reproduce. The RCT arm also has no placebo response or natural course.

The compendium's own documents had already recorded parts of this (`compendium-mathematics.tex`, Section "The treatment main-effect track", notes of 2026-10-08), and the remediation plan (`analysis/report/04-treatment-main-effect/00-remediati` Problems 4 and 5) asked for a correct small-sample test and an estimand defined under the Gompertz process. Neither can be repaired inside the current DGP, because the DGP lacks the features that make the estimand nontrivial. Hence a rebuild rather than a revision.

2. Aims (A)

Primary aim. Estimate the size and power of tests of the drug main effect β_D in the Hybrid design under the refined process, and compare them with the OL+BDC design, the two-period crossover (CO) and a parallel-group randomized controlled trial (RCT), all simulated from the same component model at the same total N.

Secondary aims.

1. Estimate the participant-by-treatment variance τ^2, and assess how well each design recovers individual drug effects.
2. Recommend an analysis for the drug main effect in each design, by the working decision rule of Section 6.4.
3. Quantify, in closed form first, how much of each design's power comes from randomized between-path contrasts and how much from modeled within-participant contrasts, and how much bias the drug-expectancy confounding introduces into each analysis.

What the study does not aim at. The biomarker interaction (papers 01, 02 and the stress test of `docs/46`); optimal design (`docs/39`); the comparison of response-curve families (paper 07).

3. Data-generating mechanisms (D)

3.1 The refined process

The base DGP is the refined process of paper 01 (Table `tab:spec`; Appendix B), with the biomarker uncoupled ($c_{bm} = 0$) in the base cell. For participant i on path p,

$$Y_{i0} = \mathrm{BL}_i, \qquad Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it}) - v_i\, q_{it} + \varepsilon^{\mathrm{me}}_{it}, \quad t \geq 1.$$

- **Components and means.** $E(TV_{it}) = G(w_t; m_{TV}, 5, 0.35)$ on calendar time; $E(PB_{it}) = \eta_t\, G(t_{pb}; m_{PB}, 5, 0.35)$; $E(BR_{it}) = G(t_{od}; m_{BR}, 5, 0.42)$ on drug, with the published ceilings $m_{TV} = m_{PB} = 6.506$ and $m_{BR} = 10.986$ unless a cell varies them. BL_i has mean 83.07 and SD 18.48 and is independent of the components (`docs/40`, Section 3).
- **Expectancy.** η_t scales both the mean and the SD of PB (SD $10\eta_t$); this is the only route by which blinding enters the data (`docs/40`, Section 3).
- **Covariance.** Separable AR(1), $K \otimes A$ with $A_{ts} = \rho^{|w_t - w_s|}$, $\rho = 0.7$ per week, and cross-component correlation $c_1 \rho^{|w_t - w_s|}$, $c_1 = 0.2$; SDs 10 (TV), $10\eta_t$ (PB) and 8 (BR) at every visit, on and off drug (paper 01, "Why separability").
- **Anchored carryover.** Off drug after exposure, $E(BR_{it}) = \mu_{BR,\mathrm{last}}\, 2^{-t_{sd}/t_{1/2}}$, decayed once from the last on-drug mean, not by the published compounding recursion (`docs/53`; paper 01, Appendix B.3). The same decay applies to the drug-response profile q_{it}.
- **Half-life.** Pharmacological range $t_{1/2} \in \{0.1, 0.2\}$ week (17 to 34 hours, about 7 to 15 plasma half-lives of prazosin; paper 01, "Pharmacological grounding"), base 0.2. A **behavioral carryover scenario** at $t_{1/2} = 1$ week is simulated and labeled as such everywhere it is reported: it is not a pharmacological claim about prazosin.
- **Re-exposure.** As in the published process, t_{od} restarts at zero when the drug is resumed (justified by re-titration of prazosin). Combined with anchored carryover, this gives a downward step in the mean at the first re-exposure visit when the residual exceeds the restarted curve (paper 01, Appendix B.3).
- **Validity.** The generator stops on a correlation matrix that is not positive definite (paper 01, Appendix B.2). With $c_{bm} = 0$ no cell is near a ceiling; the stop is kept for the cell with $c_{bm} > 0$ (Section 3.6).
- **Allocation.** Each participant is assigned to a path, or to an RCT arm, uniformly at random in every trial, redrawn if a path or arm receives fewer than 2 participants (the rule of `11-strawman-stress-test.R`, `RERUN-anchored.md`, Section 1). N is the total (NOTATION.md, rule 4).

3.2 New element: a random drug-effect slope

The refined process has no participant-by-treatment variance: the BR component varies between participants with SD 8 equally on and off drug, so its person-level part cancels in an on-minus-off contrast (`docs/51`, Feature 8; derived). Treatment-effect heterogeneity is added as a participant-specific ceiling,

$$v_i \sim N(0, \tau^2), \quad \text{independent of everything else,} \qquad q_{it} = E(BR_{it})/m_{BR},$$

that is, q_{it} is the drug-response mean of participant i's path computed with unit ceiling, including onset and anchored carryover; it does not depend on m_{BR} and is defined when $m_{BR} = 0$. Participant i's expected drug response is then $(m_{BR}+v_i)\, q_{it}$: heterogeneity follows the drug effect, is zero before exposure and decays with the drug effect, as `docs/51` recommends ("let the drug-response variance follow the drug effect"). τ is set in CAPS points, independent of m_{BR}, so that heterogeneous nulls ($m_{BR} = 0$, $\tau > 0$) exist. A normal v_i allows a participant to be harmed when $\tau > m_{BR}$; that is intended in the heterogeneous null. Whether v_i should instead scale the binary on-drug state, or be correlated with BL_i, is an open decision (Section 8, item 9).

3.3 Designs

Table 2: Designs, randomization paths, schedules and expectancy

Design	P	Visit weeks (after baseline)	η_t	Paths (drug state by visit)
Hybrid	4	4, 8, 9, 10, 11, 12, 16, 20	1, 1, then 0.5	11110010, 11110001, 11100010, 11100001
OL+BDC	2	4, 8, 12, 16, 17, 18, 19, 20	1 to week 16, then 0.5	11111100, 11111000
CO	2	every 2.5 weeks to week 20	0.5	11110000, 00001111
RCT (parallel)	2 arms	4, 8, 9, 10, 11, 12, 16, 20	0.5	11111111 or 00000000
Multi-crossover (optional)	random	eight weekly visits	0.5	random 4 + 4 per participant

The first three rows are the designs of `analysis/scripts/quick-sim/hendrickson-problems/04-power-simulation.R`, lines 74 to 83 (inspected). All have 8 post-baseline visits over 20 weeks [4].

The RCT arm. A parallel-group comparator of the kind that motivated the Hybrid design [13], double-blind throughout ($\eta_t = 0.5$ from week 0), so PB and TV follow the same curves as in the N-of-1 designs and are identical in the two arms. The active arm has $t_{od} = w_t$; the placebo arm has BR mean 0. Two properties need stating:

- **Inherited BR noise.** Under the refined process the placebo arm still carries BR noise with SD 8, as an unexposed crossover path does. This is the convention of the published construct, not pharmacology (`docs/51`, Feature 8); it adds 64 to the variance of every RCT measurement. Whether to keep it is an open decision (Section 8, item 10).
- **Baseline measured with error.** The analysis sees $Y^*_{i0} = \mathrm{BL}_i + \varepsilon_{i0}$, $\varepsilon_{i0} \sim N(0, \sigma^2_{\varepsilon,0})$, at reliability r_{BL}. Published test-retest reliabilities of the CAPS are high but below 1 [14], so an error-free baseline is the optimistic limit. The same observed baseline is used in the N-of-1 designs, where the baseline row then has its own noise, which also removes the degenerate noise-free baseline row noted in `docs/41`, Proposition 5. The default and the grid are an open decision (Section 8, item 7).

The multi-crossover. Under the refined process with re-titration, a single on-drug week reaches 3.1% of the ceiling (0.34 points at the published m_{BR}; verified, arithmetic), so the design of the current paper cannot express the drug effect of a titrated drug except through runs of consecutive on-drug weeks. It is offered only as an optional bridge to the current paper (Section 8, item 5).

3.4 Factors and grids

Table 3: Data-generating factors, levels and base values

Factor	Levels	Base
Drug effect $f_m = m_{BR}/10.986$	0, 0.2, 0.4, 0.6, 0.8, 1.0	curve; 0.4 in stress cells
Heterogeneity τ (points)	0, 2.75, 5.49 ($\tau/10.986$ = 0, 0.25, 0.5)	0
Total N	35, 70, 150	70
Carryover $t_{1/2}$ (weeks)	0, 0.1, 0.2 (pharmacological); 1.0 (behavioral, labeled)	0.2

Factor	Levels	Base
Decay form	exponential; Weibull shape 0.5 and 2 at $t_{1/2} = 1$	exponential
Natural course m_{TV}	$\times 0$, $\times 1$, $\times 2$	$\times 1$ (6.506)
Placebo response m_{PB}	$\times 0$, $\times 1$, $\times 2$	$\times 1$ (6.506)
Dropout	none; 20% monotone MCAR; 20% monotone MAR on $Y_{i,t-1}$	none
Phase measurement error	none; $\sigma_{\varepsilon,\mathrm{OL}} = 3$, $\sigma_{\varepsilon,\mathrm{bl}} = 6$	none
Baseline reliability r_{BL}	1; 0.8 ($\sigma_{\varepsilon,0} = 9.24$)	open (Section 8)

The dropout and measurement-error mechanisms are those of the options patch and the stress test (`analysis/patches/orig-to-refined/ README.md`, options table; `docs/46`, Section 5).

3.5 Effect size in outcome and SD units

The drug-effect axis is defined on the DGP, by the ceiling m_{BR}, because each design's estimand takes a different value for the same drug (Section 4). For an SD scale we propose the SD of a blinded post-baseline score conditional on baseline,

$$\sigma^2_{Y|\mathrm{BL}} = \sigma^2_{TV} + (10\eta_t)^2 + \sigma^2_{BR} + 2c_1(10\sigma_{TV}\eta_t + \sigma_{TV}\sigma_{BR} + 10\eta_t\sigma_{BR}) = 257$$

at $\eta_t = 0.5$, so $\sigma_{Y|\mathrm{BL}} = 16.03$ (derived; verified, arithmetic; the same value as `docs/41`, Proposition 3, because marginal variances do not depend on the serial structure). The unconditional SD is $\sqrt{598.5} = 24.46$.

Table 4: Drug-effect grid in multiplier, outcome and SD units

f_m	0	0.2	0.4	0.6	0.8	1.0
m_{BR} (points)	0	2.20	4.39	6.59	8.79	10.99
$m_{BR}/\sigma_{Y\|\mathrm{BL}}$	0	0.14	0.27	0.41	0.55	0.69

The grid was chosen to keep the N-of-1 designs off the power ceiling. That rests on rough derived values (an RCT SE near 3.8 points at $N = 70$ with an error-free baseline, Section 5.1; a Hybrid SE near 1 point, taken from the compound-symmetry results of `docs/40`, Section 9) and must be checked against the closed form of Section 5.1 before the run.

3.6 Draft cell table

Every block varies one factor from the base cell (Hybrid, OL+BDC, CO and RCT; refined process; $N = 70$; $t_{1/2} = 0.2$; $\tau = 0$; no dropout or measurement error). Stress blocks have a null twin ($f_m = 0$) and an alternative at $f_m = 0.4$. The RCT is not simulated in blocks whose factor does not reach it.

Table 5: Draft cell table: blocks, levels and cell counts

Block	Factor varied	Levels	f_m	Designs	Cells (null / alt)
1	power curves	base	0 to 1.0	4	24 (4 / 20)
2	heterogeneity	$\tau = 2.75$, 5.49	0, 0.4	4	16 (8 / 8)
3	sample size	$N = 35$, 150	0, 0.4	4	16 (8 / 8)
4	carryover	$t_{1/2} = 0$; 1.0; Weibull 0.5 and 2 at 1.0	0, 0.4	3 (no RCT)	24 (12 / 12)
5	natural course	$m_{TV} \times 0$, $\times 2$	0, 0.4	4	16 (8 / 8)
6	placebo response	$m_{PB} \times 0$, $\times 2$	0, 0.4	4	16 (8 / 8)
7	dropout	MCAR, MAR (20%)	0, 0.4	4	16 (8 / 8)
8	phase measure-ment error	3 / 6	0, 0.4	4	8 (4 / 4)
9	baseline error	$r_{BL} = 0.8$	0, 0.4	4	8 (4 / 4)
Total					144 (64 / 80)

Optional cells, not counted above: $t_{1/2} = 0.1$ (expected to be indistinguishable from 0.2 at weekly visits); explained heterogeneity through covariance moderation, $c_{bm} = 0.30$, $\tau = 0$ (the biomarker then induces a drug-effect SD of $c_{bm}\sigma_{BR} = 2.4$ points on drug; derived from `docs/40`, Section 5); and the multi-crossover in blocks 1 and 4.

Misspecified analysis half-life needs no new data: in the cells of block 1 and the $t_{1/2} = 1$ cells of block 4, the D_{bc}-based analyses are also fitted at $t^a_{1/2} \in \{0, 0.5, 1\}$.

Replicates and MCSE targets. Two options (Section 8, item 6):

Table 6: Replicate options with Monte Carlo standard error targets

Option	Null cells	Alternative cells	MCSE of size at 0.05	Largest MCSE of power	MCSE of coverage at 0.95
R1	2,000	1,000	0.0049	0.016	0.0049 (null), 0.0069 (alt)
R2 (as `docs/46`)	1,000	500	0.0069	0.022	0.0069 (null), 0.0097 (alt)

R1 gives 128,000 null and 80,000 alternative trials (208,000), of which about 165,000 are N-of-1 trials; R2 halves both. The MCSE of bias is $\mathrm{SD}(\widehat{\beta}_D)/\sqrt{n_{\mathrm{reps}}}$, about 0.03 points for an empirical SD of 1 at $n_{\mathrm{reps}} = 1{,}000$.

4. Estimands (E)

4.1 Candidate definitions of β_D

Under the refined process the drug effect is not one number: it rises on the Gompertz curve, restarts at re-exposure and decays after discontinuation, and the open-label visits add full expectancy. Every analysis with a single drug coefficient targets some weighted average of a drug response that ranges from 4.3 to 10.2 points across the Hybrid visits (`docs/40`, Section 9). Three definitions were considered. Values are at the published ceiling, $t_{1/2} = 0$, equal path weights (verified, arithmetic, `estimand-check.R`):

1. **Full-effect contrast**, $\beta_D^{\mathrm{full}} = -m_{BR}$: the reduction after long exposure, 10.99 points. Design-free, but no design observes it: no Hybrid or crossover visit is past 10 weeks of continuous exposure, and the re-exposure visits reach 39% of the ceiling. Only an analysis that knows the Gompertz shape can target it (the oracle analysis OP, Section 5); every other analysis is biased toward zero by construction, which makes bias uninformative about the analyses.
2. **Contrast at the design's time on drug**, $\beta_D^{\mathrm{exp}} = -[\overline{E(BR)}_{\mathrm{on}} - \overline{E(BR)}_{\mathrm{off}}]$, the on-minus-off difference in the mean drug response over the design's own visits, averaged over participants, net of PB and TV by definition. It is what the drug did at the times the design exposed people.

Values: Hybrid −7.22 (paths 1 and 2, −7.55; paths 3 and 4, −6.89); OL+BDC −9.35; CO −6.70; RCT at week 20, −10.97. Under carryover the off-drug mean is positive and the value shrinks; under expectancy the definition excludes PB, so an analysis that credits open-label expectancy to the drug is biased away from zero.

3. **Blinded on-minus-off contrast at matched expectancy**, β_D^{bl}: the same contrast restricted to blinded visits. Hybrid −7.56 (paths 1 and 2, −8.09; 3 and 4, −7.04); undefined for OL+BDC except at its two or three blinded visits. Its randomized version, the equal-weight average of the between-path contrasts at weeks 10, 16 and 20, is −6.25 in the Hybrid design, the probability limit of the visit-factor analysis (`docs/40`, Section 9; `compendium-mathematics.tex`, Equation `eq:bp-fwl`). Matching expectancy removes the PB step but not the natural course: the blinded within-path contrast without time adjustment has a null bias of +0.37 points, against +1.29 for the contrast over all visits (verified, arithmetic).

Implications under Gompertz onset. Definitions 2 and 3 depend on the schedule, so designs differ in their estimand for the same drug: the RCT's week-20 contrast is 1.5 times the Hybrid's β_D^{exp}. That difference is a real property of the designs (short re-exposures of a titrated drug), and it will appear in power. It means that power can be compared across designs at a common m_{BR}, but bias and coverage only against each design's own estimand value.

Implications under expectancy. For definition 2, the limits of `docs/40`, Section 9, at $t_{1/2} = 0$ are linear time −8.21, linear time with η_t −7.28, spline −6.22, visit factor −6.25, all under working independence and the published allocation. Against $\beta_D^{\mathrm{exp}} = -7.22$ linear time overstates the effect by about 1 point and the expectancy model is close (inferred from those limits and the value above; the weightings differ slightly, so this must be recomputed in the closed-form step).

The test is common to all definitions. Under $m_{BR} = 0$ every definition is zero, so size and power refer to the same null hypothesis in every design. The estimand choice affects bias, coverage and the interpretation of $\hat{\beta}_D$, not the rejection rates.

4.2 Recommendation (open decision)

We recommend **definition 2**, β_D^{exp}, as the primary estimand for estimation, with definition 3's randomized version reported as a secondary estimand for the Hybrid design. Reasons: it is defined for every design including the RCT; it is a pharmacological quantity, so expectancy confounding shows as bias rather than being built into the target; and it is attainable by analyses a trial can run. Its cost is that it is design-specific. This choice is open (Section 8, item 1). Its value in every cell is computed from the population means of the cell, in closed form, and stored with the cell.

4.3 Secondary estimands

- **τ^2, on the scale of the primary estimand.** Participant i's exposure-averaged effect is $-(m_{BR} + v_i)\,\bar{q}_d$, with $\bar{q}_d$ the design's on-minus-off mean of q_{it}; its variance is $\tau_d^2 = \tau^2 \bar{q}_d^2$ (derived). The random-slope analyses of Section 5 target τ_d^2, not τ^2; both are reported.
- **Individual effects.** The participant-level value $-(m_{BR} + v_i)\,\bar{q}_{d,p(i)}$, conditional on v_i and excluding the transient BR noise.
- **Size.** The rejection rate under $m_{BR} = 0$, with and without heterogeneity.

5. Methods (M)

5.1 Closed form first

Before any simulation, every analysis whose mean model is fixed is evaluated in closed form, extending `docs/37` and `docs/39` from the biomarker slope to the drug coefficient.

- **Probability limit.** For an analysis with mean design X_p on path p and working covariance V, $\operatorname{plim} \hat{\beta}_D = \left[\left(\sum_p \varpi_p X_p' V^{-1} X_p\right)^{-1} \sum_p \varpi_p X_p' V^{-1} \mu_p\right]_D$, with $\varpi_p = 1/P$ the expected path share (local symbol; π is reserved for power) and μ_p the true mean vector (`compendium-mathematics.tex`, "Time adjustment and the probability limit of the drug coefficient", extended from working independence to a general V; derived). Its difference from the estimand value is the bias, and under $m_{BR} = 0$ a nonzero limit predicts size inflation. Linear time without η_t is predicted to have a nonzero limit under the null in the Hybrid and OL+BDC designs (inferred from the 2-point difference between linear time and visit effects in `docs/40`, Section 9).
- **Variance.** The sandwich $\operatorname{Var}(\hat{\beta}_D) = N^{-1}[B^{-1}(\sum_p \varpi_p X_p' V^{-1} \Sigma_p V^{-1} X_p) B^{-1}]_{DD}$, $B = \sum_p \varpi_p X_p' V^{-1} X_p$, with Σ_p the true covariance of $\mathbf{Y}_i$: the refined separable covariance plus BL variance, plus $\tau^2 \mathbf{q}_p \mathbf{q}_p'$, plus measurement error. For the summary-measure analyses PW and PR this is exact (the contrast is a fixed linear combination, as E9 is in `docs/37`, Section 4); for the likelihood analyses it holds at the probability limit of the working covariance parameters, which must be computed by fitting the working model to Σ_p (approximate).
- **Contrast variance by design.** `docs/39`, Section 3, gives the contrast variances of the on-minus-off weights by design under CS and AR(1) (Hybrid 107.3 to 126.8 under AR(1) at $\rho = 0.8$ without measurement error, against 212.8 for CO; to be recomputed at the refined $\rho = 0.7$). For the main effect the same quadratic form $a'Ca$ applies, with two changes: the drug signal is $a'\mu_{BR}$ instead of the coupling term, and the weights must also remove the non-drug mean, either by orthogonality to the time and

expectancy terms or by the between-path construction. The heterogeneity adds $\tau^2(a'q)^2$.

- **RCT.** For ANCOVA at week 20 with r_{BL}, $\mathrm{Var}(\hat{\beta}_D) \approx (2/N)(2\sigma^2_{20|Y_0^*} + \tau^2 q^2_{20})$ at equal arms, with $\sigma^2_{20|Y_0^*} = 598.5 - 18.48^2\, r_{BL}$ the placebo-arm residual variance (derived). At $N = 70$ and $\tau = 0$: SE 3.83 at $r_{BL} = 1$, 4.08 at 0.9, 4.31 at 0.8; normal-approximation power at the published ceiling 0.82, 0.77, 0.72 (verified, arithmetic; ignores the t reference distribution).
- **Analytic power curves.** Power by noncentral t at the analysis's reference degrees of freedom, over the f_m grid, for every design. The simulation then validates the closed form in the cells of block 1 and of blocks 2 and 4: agreement within 2 MCSE for rejection rates and within 2 MCSE for the empirical SD, as `docs/37`, Section 6 and `docs/39`, Section 9 required. Disagreement is reported, not tuned away.

The closed form is also the gate for the grids: if it shows a design at power above 0.99 or below 0.10 across most of the grid, the grid is revised before the run, and the revision is recorded.

5.2 Analyses

The analyses adapt the fourteen stress-test analyses (`compendium-mathematics.tex`, Table `tab:invd-analyses`; `docs/47`) to the main effect by dropping the biomarker terms, and add analyses that only the main effect needs. IDs shared with the stress test denote the main-effect analog of the same model. Notation: X_{it} is $D_{bc,it}$ at $t^a_{1/2}$, with the pre-exposure guard (0 at baseline and before first exposure; paper 01, Appendix B.4), and $X_{it} = D_{it}$ when $t^a_{1/2} = 0$. Default $t^a_{1/2} = 0.2$ week. All N-of-1 analyses use the long data with the observed baseline as a row at $t = 0$ unless stated. RC is `nlme::lme` with a random intercept and `corCAR1(form = ~ t | ptID)` on weeks, REML; RCH adds `varIdent(form = ~ 1 | phase)` (baseline, open label, blinded); CR2 is `clubSandwich::coef_test(vcov = 'CR2', test = 'Satterthwaite')` clustered on participant [5, 6].

Table 7: Analyses of the drug main effect: mean model, covariance, inference, degrees of freedom and purpose

ID	Mean model	Covariance	Inference	df	Purpose
S	$t + X$	RC	CR2	Satterthwaite (CR2)	stress-test reference; exposes the expectancy confounding
F1	$t + X + \eta_t$	RC	CR2	Satterthwaite (CR2)	expectancy-term model; proposed reference (strawman)
G1	as F1	RCH	CR2	Satterthwaite (CR2)	phase variances
G2	as F1	RCH	model-based	containment	phase variances, model-based
NS	`ns(t, 3)` $+X + \eta_t$	RC	CR2	Satterthwaite (CR2)	flexible time (`docs/42`)
VF	visit factor $+X$	RC	CR2	Satterthwaite (CR2)	randomized between-path contrasts only
A	$t + X$	unstructured, `mmrm`	Kenward-Roger	Kenward-Roger	stress-test analog; known liberal at small N
VA	visit factor $+X$	unstructured, `mmrm`	Kenward-Roger	Kenward-Roger	the MMRM convention [7]
R	$t + D$	compound symmetry, `mmrm`	model-based	Satterthwaite (`mmrm`)	published-type reference [4]

ID	Mean model	Covariance	Inference	df	Purpose
XS	$t + D + \eta_t$, first off-drug visit after each discontinuation dropped by schedule	RC	CR2	Satterthwaite (CR2)	period exclusion without oracle information; versus D_{bc}
XO	as XS, dropping visits whose true carryover fraction exceeds 0.05	RC	CR2	Satterthwaite (CR2)	oracle; bridge to the v1 analysis
PW	per participant, blinded on-minus-off mean; one-sample t	none	exact t	$N - 1$	E9 analog; time-confounded by construction
PR	per participant, $s_i = \sum_t (D_{p(i)t} - \bar{D}_t) Y_{it}$; path-stratified mean	none	t	$N - P$	randomization-based summary measure (the FWL estimator)
RS	as F1, random slope on X correlated with the intercept	RC + slope	CR2	Satterthwaite (CR2)	τ^2, individual effects
RSL	as RS without η_t	RC + slope	CR2	Satterthwaite (CR2)	τ^2 under linear time

ID	Mean model	Covariance	Inference	df	Purpose
OP	$t + \eta_t + Q$, $Q_{it} = q_{it}$ the true profile	RC	CR2	Satterthwaite (CR2)	oracle shape; targets β_D^{full}
AN	RCT: week-20 score on arm and Y_{i0}^*	OLS	HC2	$N - 3$	RCT primary
AN0	as AN on the error-free BL_i	OLS	HC2	$N - 3$	oracle; reproduces the v1 RCT
AM	RCT: all visits, visit-by-arm, Y_{i0}^* covariate, contrast at week 20	unstructured, `mmrm`	Kenward-Roger	Kenward-Roger	RCT using every visit

Notes on the table:

- **Kenward-Roger** is the `mmrm` implementation. The stress test used its linear variant (`Kenward-Roger-Linear`; compendium, "Inference"); the same variant is used here for comparability, which should be confirmed (Section 8, item 4) [8].
- **CR2 as default.** CR2 with Satterthwaite degrees of freedom is the default inference for every random-intercept model, because model- based and Kenward-Roger inference were anticonservative with flexible time in the Hybrid carryover cell and CR2 was not (`docs/42`, update of 2026-10-07; `docs/45`, Section 4.4), and A failed the size screen of the stress test (0.087 at $N = 35$; `docs/47`, Section 3).
- **HC2** for the RCT is the singleton-cluster case of CR2; it protects against the unequal arm variances that $\tau > 0$ creates.
- **F1 in CO.** With η_t constant after baseline, η_t is a baseline indicator there, so F1 frees the baseline mean.
- **XS against D_{bc}.** XS replaces the v1 oracle exclusion by a rule a trial can apply. In the Hybrid design it drops weeks 11 (paths 1 and 2) or 10 (paths 3 and 4) and the first post-crossover off-drug visit.
- **PW and PR** are the main-effect counterparts of E9. PW keeps the time confounding of a within-path contrast (null bias +0.37 points over blinded

visits; Section 4.1) and is included because its moments are exact and it shows the size of the confounding. PR uses only the visits at which paths differ in drug state, so its mean is the randomized contrast and its variance is exact.

- **Random slopes.** In the Hybrid design each participant has four or five on-drug and three or four off-drug visits, so the slope variance may be weakly identified and hit the boundary. τ_d^2 is tested by the likelihood ratio against RC with the 50:50 mixture of χ_1^2 and χ_2^2 [9]; its interval is the `nlme::intervals()` approximation (Wald on the log SD scale), whose coverage is itself a result. Boundary and nonconvergence rates are reported.
- **Misspecified half-life.** S, F1, RS and XS are refitted at $t_{1/2}^a \in \{0, 0.5, 1\}$ in the cells named in Section 3.6. The AIC-selected half-life is not included: it failed the size screen with the expectancy terms in the stress test (F1A, 0.057 pooled; `docs/47`, Section 8).

6. Performance measures (P)

6.1 Measures

For each analysis and cell, with MCSEs as in Morris, White and Crowther [1], Table 6:

Table 8: Performance measures and their Monte Carlo standard errors

Measure	Definition	MCSE
Size (null cells)	$\hat{\pi}_0 = n_{\text{reps}}^{-1} \sum I(p_r < 0.05)$	$\sqrt{\hat{\pi}_0(1-\hat{\pi}_0)/n_{\text{reps}}}$
Power	as above, alternative cells	as above
Size-adjusted power	rejection at the 5% quantile of the analysis's null p-values in the matching null cell	binomial, conditional on the quantile
Bias	$\bar{\beta} - \beta_D$, against the cell's estimand value	$\text{SD}(\hat{\beta})/\sqrt{n_{\text{reps}}}$
Empirical SE	$\text{SD}(\hat{\beta})$	$\text{SD}(\hat{\beta})/\sqrt{2(n_{\text{reps}}-1)}$
Model SE calibration	$\kappa = \overline{\text{SE}}/\text{SD}(\hat{\beta})$	delta method [1]
Coverage	share of nominal 95% intervals containing β_D	binomial
τ_d^2 bias, coverage	as above for the RS and RSL estimates and intervals	as above

Measure	Definition	MCSE
Individual effects	RMSE and correlation of the BLUP with the true individual value	bootstrap over replicates
Failed fits, boundary fits	rates	binomial

Failed fits are excluded from rates and counted, as in the stress test (`docs/46`, Amendment 2, item 3); an analysis with more than 2% failed fits in any cell is flagged.

6.2 The nominal band

At R1, a single null cell has MCSE 0.0049 and a 2-MCSE band of 0.040 to 0.060; the pooled rate over the 51 N-of-1 null cells of one analysis across designs has MCSE about 0.0007. At R2 the single-cell band is 0.036 to 0.064, as in `docs/46`.

6.3 Paired comparisons

All analyses of a design are fitted to the same trials, and power differences within a cell are tested by McNemar's test. Designs are not paired: their schedules differ. Participant-level draws that do not depend on the schedule (BL_i, v_i, ε_{i0}) use common random numbers across designs; design comparisons are otherwise treated as independent proportions.

6.4 Working decision rule

Within each design, adapted from `docs/46`, Section 7, and stated as a working rule that may be revised with recorded reasons:

1. **Size screen.** Discard an analysis whose pooled null rejection rate over that design's null cells lies outside 2 MCSE of 0.05, or that exceeds 0.075 in any null cell. Heterogeneous nulls (block 2) count; they test whether ignoring τ^2 costs size.
2. **Estimation screen** (new). In the base alternative cells, flag an analysis whose coverage of β_D is below 0.925 or whose absolute bias exceeds 10% of $|\beta_D|$. Flagged analyses may still be reported for testing, not for estimation.
3. **Oracle exclusion.** XO, OP and AN0 are never selected.
4. **Selection.** Among survivors, select by minimax regret: the smallest largest shortfall in size-adjusted power over the alternative cells.
5. **Margin.** The proposed reference F1 is retained unless a survivor beats it by more than 0.05 in largest shortfall, with the McNemar difference significant at 0.05 in at least half the alternative cells. Ties within 0.02 go to the analysis with fewer covariance parameters.

Across designs. Power curves of each design's selected analysis are compared at common m_{BR} and N; differences are tested as independent proportions. No design is "selected": the comparison reports operating characteristics.

7. Implementation plan

7.1 What to reuse

- **The stress-test driver** `analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stres` (inspected): its `build()` already gives anchored carryover and randomized allocation by default (lines 25 to 35, 118 to 133), the separable AR(1) construct, Weibull decay, dropout, measurement error, per-chunk seeds and checkpointing. It is the basis of the new driver.
- **The patches** `analysis/patches/orig-to-refined/` implement the refined DGP and fifteen analyses selectable by `op$analysis`, verified against the driver. They are the reference for reproducing the refined process in Hendrickson's code, but not the engine: their `generateSimulatedResults()` fits the analysis once to the whole population, which its README reports as impractical with the refined analyses (a single F1 fit on 7,200 participants did not finish in 10 minutes), and their analyses test `bmc:Dbc` only.
- **Design definitions** from `hendrickson-problems/04-power-simulation.R` (lines 74 to 97).

7.2 What to add

1. **Closed-form module** (`01-closed-form.R`): estimand values per cell; probability limits and sandwich variances for every fixed- mean analysis; analytic power curves. Runs first; gates the grid.
2. **Drug-effect scale.** The BR, TV and PB ceilings are hard-coded in `build()` (10.98604 at line 120; 6.50647 at lines 134 and 135); make them cell parameters.
3. **Random drug slope.** $v_i\, q_{it}$ added after the draw, with q_{it} from the unit-ceiling profile.
4. **Designs.** OL+BDC and CO in the driver (11 uses Hybrid only, line 72); the RCT arm; the optional multi-crossover.
5. **Observed baseline with error** for all designs.
6. **Main-effect analyses** of Section 5.2, each returning the estimate, SE, df, p, interval, and for RS and RSL the variance component, its test and the BLUPs.
7. **Unit checks** before the run: the estimand values reproduce Section 4.1 at the published ceiling; the RCT closed form matches a large draw; τ^2 is recovered from the generated individual effects; allocation shares; the closed form against a direct Monte Carlo of PW and PR.

7.3 Compute estimate (inferred)

From `RERUN-anchored.md`, Section 3: the stress test costs about 11 worker-seconds per trial for about twelve analyses, dominated by three unstructured Kenward-Roger fits; its post hoc runs cost about 0.4 worker-seconds per random-intercept fit with CR2. The list above has nine random-intercept fits, two unstructured fits, one `mmrm` compound symmetry fit, two random-slope fits and two summary measures, about 11 worker-seconds per N-of-1 trial if a random-slope fit costs 1 second (assumed). At R1 that is about 165,000 N-of-1 trials, 500 worker-hours: about 63 hours on 8 cores, 170 hours at simrun's default of 3 cores. R2 halves this. The $N = 150$ cells add perhaps 10% (assumed). RCT fits are negligible. A pilot of 20 replicates per cell is run first to time the fits, as in `docs/46`, Section 9.

7.4 Execution on the simulation server

Through `simrun`, as described in `RERUN-anchored.md`, Section 4.2: the job runs the committed tree, so the driver must be committed first, and `simrun submit` refuses while tracked files have uncommitted changes; commands are argument vectors without redirection; the job's `SIM_CORES` sets the worker count; results come back with `simrun fetch` and `simrun apply`. One job per block keeps jobs short and restartable. The core count the host can give is unknown (unverified).

7.5 Output naming

- Drivers: `analysis/scripts/treatment-main-effect/rebuild/`, `01-closed-form.R`, `02-simulate.R`, `03-decision.R`, `04-tables-figures.R`. The v1 drivers stay where they are.
- Data: `analysis/data/treatment-main-effect/rebuild/`: `closed-form-{estimands,plim,power}.c` `cells-{tag}/` with chunk files; `replicates-{tag}.rds`; `summary-{tag}.csv`; `decision-{screen,estimation,power,minimax}-{tag}.csv`. Tags `full`, `pilot20`, and a block suffix (`-b1` to `-b9`) for server jobs.
- Seeds per chunk, $20261009 + 1000 \times \text{cell} + \text{chunk}$, so results do not depend on the number of cores (the convention of `docs/47`, Section 11).

8. Open decisions for the user

1. **Estimand definition.** Adopt β_D^{exp} (exposure- averaged on-minus-off drug response at the design's own visits) as primary, with the randomized blinded contrast as secondary for the Hybrid design, or choose the full-effect or the blinded contrast as primary (Section 4).
2. τ^2 **grid.** $\tau \in \{0, 2.75, 5.49\}$ points (0, 0.25 and 0.5 of the published ceiling), set in absolute units so that heterogeneous nulls exist; or a grid relative to m_{BR}.

3. **Effect-size grid units.** The multiplier f_m on the drug-response ceiling, reported in points and in units of $\sigma_{Y|\mathrm{BL}} = 16.03$; or another SD (the unconditional 24.46, or an RCT-based Cohen's d).
4. **Analysis list.** The nineteen analyses of Section 5.2, including the oracle arms XO, OP and AN0, the variant of Kenward-Roger, and the default $t^a_{1/2} = 0.2$ week; drop or add analyses.
5. **Multi-crossover.** Whether to keep the current paper's design as an additional design (bridge to v1, expected to have little power under re-titration), and if so in which blocks.
6. **Replicate counts.** R1 (2,000 null, 1,000 alternative; about 63 hours on 8 cores) or R2 (1,000 and 500, as `docs/46`; about 32 hours).
7. **RCT baseline.** Whether the RCT is analyzed by ANCOVA on a baseline measured with error, and how much error: reliability 1 (error-free, as v1), 0.9 or 0.8; and whether the same observed baseline is used in the N-of-1 designs (proposed: yes).

Further decisions that the protocol has had to assume:

8. **Base half-life** 0.2 week, with 0.1 optional and 1 week as the labeled behavioral scenario.
9. **Form of heterogeneity.** v_i scaling the Gompertz profile (proposed), or scaling the binary drug state; independent of BL_i (proposed) or correlated with it.
10. *BR* **noise in unexposed participants.** Keep the inherited SD 8 off drug and in the RCT placebo arm (the refined process as specified), or make the *BR* SD follow the drug effect.
11. **Decision rule.** The estimation screen of Section 6.4, item 2, is new relative to `docs/46`; its thresholds (coverage 0.925, bias 10%) and the single-cell cap of 0.075 at R1 (5 MCSE) need confirming.

9. What carries over from the current paper

Carried over, with edits (`report.Rmd`):

- The Introduction's background on N-of-1 designs and sample size (lines 76 to 141), including the Senn and Zucker variance-component apparatus (lines 142 to 252), which becomes directly relevant once τ^2 is in the DGP; references [2, 3, 10, 11] stay.
- The single-case framework, prazosin, regulatory and literature sections (lines 290 to 458), with the comparisons of Blackston et al. [12] and others, rewritten where they quote the paper's own results.
- The comparison with prior simulation studies in the Discussion (lines 1770 to 1922), kept as structure, with every number replaced.

Replaced. The abstract; the Objectives (line 460); all of Methods (lines 472 to 600); all Results (lines 1053 to 1573); the remaining Discussion and the Conclusions; the cover letter, title page, short report and bullets. The word "hybrid" is removed from every statement about the v1 simulation.

Bibliography fix found while reading. The key `araujoComparisonFourMethods2014` in the paper's `references.bib` (line 22) holds Chen and Chen (2014); the key should be renamed (inspected).

10. References

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References 2, 3, 8 to 14 were checked against PubMed records on 2026-10-08 (titles, volumes, pages, DOIs; Stram and Lee has no DOI in its record). References 1, 4 to 6 are as cited in `docs/46` and `docs/47`; reference 7 is a book, not checked. The CAPS reliability values of reference 14 were not read from the article; only the record's statement that reliability is high was checked.

Changes to the working protocol

None recorded as of 2026-10-08. Changes made after results are seen will be appended as numbered amendments, each with its date, scope and reason.